Model Predictive Control
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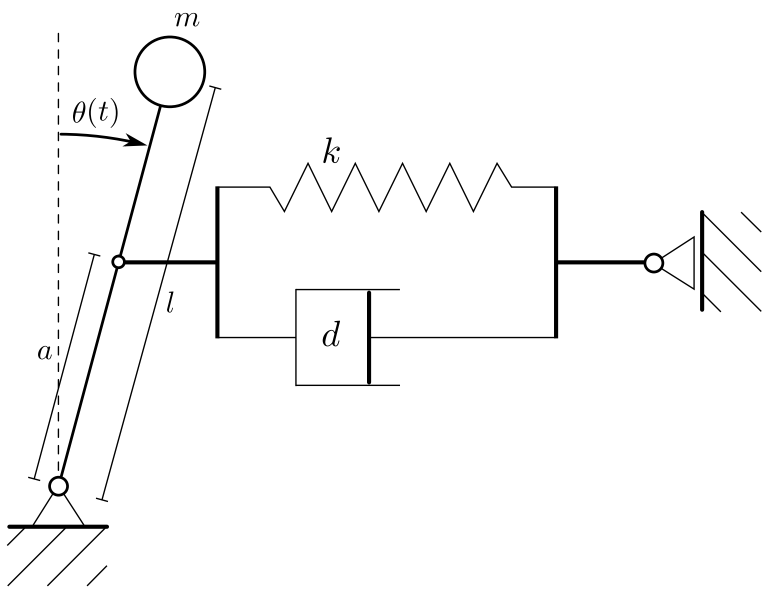
Recitation 01
System Modeling, Linearization & Discretization

Exercise 1

We consider the following problem :



- The spring is at rest at $\theta(t) = 0$.
- The resulting spring-damper force always acts horizontally on the pendulum.
- We neglect the vertical movement of the spring-damper system.

1. Find the differential equation governing $\theta(t)$ for the autonomous non-linear pendulum system in state space form.
2. Simulate the system from different initial positions $\theta(0)$, $\dot{\theta}(0)$.

Exercise 2

1. Find all equilibrium points and linearize the system at one where $\theta(t) = 0$.
2. Compare linearized and nonlinear system in simulation.

Exercise 3

1. Discretize the nonlinear system using an Euler-Forward scheme.
2. Discretize the linearized system using an Euler-Forward scheme.
3. Discretize the linearized system using Exact discretization.
4. Compare the resulting system descriptions.